Air System Sizing Summary for HP-101-FITNESS
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	HP-101-FITNESS		Number of zones	1	
Equipment Class	SPLT AHU		Floor Area	1279.5	sqft
Air System Type	SZCAV		Location	St George Municipal, UT, USA	

Location not in excel

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Not in excel. Shared with zone summary

Central Cooling Coil Sizing Data

Total coil load	3.2	Tons	Peak coil load occurs at	July 16:00	
Total coil load	38.9	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	34.5	MBH	Entering DB / WB	82.0 / 64.5	F
Coil CFM at peak load	1215	CFM	Leaving DB / WB	53.4 / 53.3	F
Sum of peak zone CFM	1215	CFM	Resulting RH	62	%
Sensible heat ratio	0.887		Design supply temp.	55.0	F
CFM/Ton	374.7		Zone T-stat Check	1 of 1	OK
sqft/Ton	394.6		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	30.4				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	21.8	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	1215	CFM	BTU/(hr sqft)	17.0	
Max coil CFM	1215	CFM	Ent. DB / Lvg DB	57.7 / 76.1	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	1215	CFM	Fan motor BHP	0.67	BHP
Design CFM/sqft	0.95	CFM/sqft	Fan motor kW	0.53	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	335	CFM	CFM/person	26.20	CFM/person
CFM/sqft	0.26	CFM/sqft			

Air System Sizing Summary for HP-102-OFFICES

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton)	05/29/2026
Prepared by: Shakespeare Engineering	1:21 PM

Air System Information

Air System Name	HP-102-OFFICES		Number of zones	1	
Equipment Class	SPLT AHU		Floor Area	774.0	sqft
Air System Type	SZCAV		Location	St George Municipal, UT, USA	

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	0.7	Tons	Peak coil load occurs at	July 16:00	
Total coil load	8.8	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	8.8	MBH	Entering DB / WB	80.2 / 61.7	F
Coil CFM at peak load	333	CFM	Leaving DB / WB	53.4 / 51.9	F
Sum of peak zone CFM	333	CFM	Resulting RH	50	%
Sensible heat ratio	1.000		Design supply temp.	55.0	F
CFM/Ton	453.1		Zone T-stat Check	1 of 1	OK
sqft/Ton	1054.7		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	11.4				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	6.4	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	333	CFM	BTU/(hr sqft)	8.3	
Max coil CFM	333	CFM	Ent. DB / Lvg DB	60.7 / 80.4	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	333	CFM	Fan motor BHP	0.18	BHP
Design CFM/sqft	0.43	CFM/sqft	Fan motor kW	0.14	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	73	CFM	CFM/person	13.83	CFM/person
CFM/sqft	0.09	CFM/sqft			

Air System Sizing Summary for HP-103-BOH EMPLOYEE

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	**HP-103-BOH EMPLOYEE**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**593.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.1**	Tons	Peak coil load occurs at	**September 15:00**	
Total coil load	**13.3**	MBH	OA DB / WB	**101.0 / 63.2**	F
Sensible coil load	**13.3**	MBH	Entering DB / WB	**76.8 / 59.3**	F
Coil CFM at peak load	**584**	CFM	Leaving DB / WB	**53.7 / 50.6**	F
Sum of peak zone CFM	**584**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**526.7**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**535.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**22.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**7.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**584**	CFM	BTU/(hr sqft)	**12.3**	
Max coil CFM	**584**	CFM	Ent. DB / Lvg DB	**64.9 / 77.7**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**584**	CFM	Fan motor BHP	**0.32**	BHP
Design CFM/sqft	**0.98**	CFM/sqft	Fan motor kW	**0.25**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**80**	CFM	CFM/person	**8.27**	CFM/person
CFM/sqft	**0.14**	CFM/sqft			

Air System Sizing Summary for HP-104-BOH LAUNDRY

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	**HP-104-BOH LAUNDRY**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1317.5**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.6**	Tons	Peak coil load occurs at	**August 15:00**	
Total coil load	**31.4**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**30.6**	MBH	Entering DB / WB	**77.8 / 62.5**	F
Coil CFM at peak load	**1293**	CFM	Leaving DB / WB	**53.9 / 53.8**	F
Sum of peak zone CFM	**1293**	CFM	Resulting RH	**55**	%
Sensible heat ratio	**0.972**		Design supply temp.	**55.0**	F
CFM/Ton	**493.5**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**502.9**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**23.9**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**14.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1293**	CFM	BTU/(hr sqft)	**10.9**	
Max coil CFM	**1293**	CFM	Ent. DB / Lvg DB	**64.5 / 75.8**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1293**	CFM	Fan motor BHP	**0.71**	BHP
Design CFM/sqft	**0.98**	CFM/sqft	Fan motor kW	**0.56**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**188**	CFM	CFM/person	**18.99**	CFM/person
CFM/sqft	**0.14**	CFM/sqft			

Air System Sizing Summary for HP-105-LOBBY/MARKET
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) — 05/29/2026
Prepared by: Shakespeare Engineering — 1:21 PM

Air System Information

Air System Name	HP-105-LOBBY/MARKET		Number of zones	1	
Equipment Class	SPLT AHU		Floor Area	1301.9	sqft
Air System Type	SZCAV		Location	St George Municipal, UT, USA	

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	2.5	Tons	Peak coil load occurs at	July 16:00	
Total coil load	29.6	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	29.0	MBH	Entering DB / WB	83.6 / 63.3	F
Coil CFM at peak load	918	CFM	Leaving DB / WB	51.7 / 51.6	F
Sum of peak zone CFM	918	CFM	Resulting RH	55	%
Sensible heat ratio	0.980		Design supply temp.	55.0	F
CFM/Ton	372.0		Zone T-stat Check	1 of 1	OK
sqft/Ton	527.7		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	22.7				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	23.4	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	918	CFM	BTU/(hr sqft)	18.0	
Max coil CFM	918	CFM	Ent. DB / Lvg DB	55.2 / 81.3	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	918	CFM	Fan motor BHP	0.50	BHP
Design CFM/sqft	0.70	CFM/sqft	Fan motor kW	0.40	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	298	CFM	CFM/person	10.17	CFM/person
CFM/sqft	0.23	CFM/sqft			

Air System Sizing Summary for HP-106-DINING LARGE

(In Alternative: Default Alternative)

Air System Information

Air System Name	**HP-106-DINING LARGE**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1092.2**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.9**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**34.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**34.6**	MBH	Entering DB / WB	**93.7 / 63.6**	F
Coil CFM at peak load	**833**	CFM	Leaving DB / WB	**51.6 / 48.0**	F
Sum of peak zone CFM	**833**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**289.0**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**379.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**31.6**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**34.5**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**833**	CFM	BTU/(hr sqft)	**31.6**	
Max coil CFM	**833**	CFM	Ent. DB / Lvg DB	**39.4 / 81.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**833**	CFM	Fan motor BHP	**0.46**	BHP
Design CFM/sqft	**0.76**	CFM/sqft	Fan motor kW	**0.36**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**524**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Air System Sizing Summary for HP-107-BUFFET/SERVING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	HP-107-BUFFET/SERVING	Number of zones	1	
Equipment Class	SPLT AHU	Floor Area	747.0	sqft
Air System Type	SZCAV	Location	St George Municipal, UT, USA	

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	1.9	Tons	Peak coil load occurs at	July 16:00	
Total coil load	23.3	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	23.3	MBH	Entering DB / WB	82.4 / 61.5	F
Coil CFM at peak load	780	CFM	Leaving DB / WB	52.1 / 50.3	F
Sum of peak zone CFM	780	CFM	Resulting RH	47	%
Sensible heat ratio	1.000		Design supply temp.	55.0	F
CFM/Ton	401.0		Zone T-stat Check	1 of 1	OK
sqft/Ton	384.3		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	31.2				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	14.9	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	780	CFM	BTU/(hr sqft)	20.0	
Max coil CFM	780	CFM	Ent. DB / Lvg DB	57.1 / 76.7	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	780	CFM	Fan motor BHP	0.43	BHP
Design CFM/sqft	1.04	CFM/sqft	Fan motor kW	0.34	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	224	CFM	CFM/person	22.50	CFM/person
CFM/sqft	0.30	CFM/sqft			

Air System Information

Air System Name	**HP-108-FOOD PREP**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**695.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**3.0**	Tons	Peak coil load occurs at	**July 16:00**	
Total coil load	**36.6**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**36.6**	MBH	Entering DB / WB	**87.8 / 61.9**	F
Coil CFM at peak load	**1080**	CFM	Leaving DB / WB	**53.5 / 49.1**	F
Sum of peak zone CFM	**1080**	CFM	Resulting RH	**42**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**354.5**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**228.3**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**52.6**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**26.6**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1080**	CFM	BTU/(hr sqft)	**38.2**	
Max coil CFM	**1080**	CFM	Ent. DB / Lvg DB	**48.6 / 73.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1080**	CFM	Fan motor BHP	**0.59**	BHP
Design CFM/sqft	**1.55**	CFM/sqft	Fan motor kW	**0.47**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**487**	CFM	CFM/person	**35.00**	CFM/person
CFM/sqft	**0.70**	CFM/sqft			

Air System Information

Air System Name	**HP-109-DINING FLEX**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**362.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.1**	Tons	Peak coil load occurs at	**August 15:00**	
Total coil load	**12.9**	MBH	OA DB / WB	**106.2 / 66.2**	F
Sensible coil load	**12.9**	MBH	Entering DB / WB	**88.9 / 62.5**	F
Coil CFM at peak load	**361**	CFM	Leaving DB / WB	**52.8 / 49.2**	F
Sum of peak zone CFM	**361**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**337.4**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**338.6**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**35.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**11.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**361**	CFM	BTU/(hr sqft)	**31.5**	
Max coil CFM	**361**	CFM	Ent. DB / Lvg DB	**47.0 / 79.4**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**361**	CFM	Fan motor BHP	**0.20**	BHP
Design CFM/sqft	**1.00**	CFM/sqft	Fan motor kW	**0.16**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**174**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Air System Sizing Summary for HP-110-DINING SMALL
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	**HP-110-DINING SMALL**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**713.0**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.0**	Tons	Peak coil load occurs at	**August 14:00**	
Total coil load	**24.5**	MBH	OA DB / WB	**104.9 / 65.8**	F
Sensible coil load	**24.5**	MBH	Entering DB / WB	**88.5 / 62.4**	F
Coil CFM at peak load	**703**	CFM	Leaving DB / WB	**53.2 / 49.3**	F
Sum of peak zone CFM	**703**	CFM	Resulting RH	**44**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**344.1**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**349.1**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**34.4**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**22.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**703**	CFM	BTU/(hr sqft)	**31.4**	
Max coil CFM	**703**	CFM	Ent. DB / Lvg DB	**46.8 / 79.5**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**703**	CFM	Fan motor BHP	**0.38**	BHP
Design CFM/sqft	**0.99**	CFM/sqft	Fan motor kW	**0.31**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**342**	CFM	CFM/person	**12.00**	CFM/person
CFM/sqft	**0.48**	CFM/sqft			

Air System Sizing Summary for HP-111-MEETING
(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	**HP-111-MEETING**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**808.8**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**2.4**	Tons	Peak coil load occurs at	**September 14:00**	
Total coil load	**29.4**	MBH	OA DB / WB	**99.7 / 62.8**	F
Sensible coil load	**28.4**	MBH	Entering DB / WB	**76.7 / 62.3**	F
Coil CFM at peak load	**1281**	CFM	Leaving DB / WB	**54.3 / 54.1**	F
Sum of peak zone CFM	**1281**	CFM	Resulting RH	**56**	%
Sensible heat ratio	**0.968**		Design supply temp.	**55.0**	F
CFM/Ton	**523.6**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**330.6**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**36.3**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**15.3**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**1281**	CFM	BTU/(hr sqft)	**18.9**	
Max coil CFM	**1281**	CFM	Ent. DB / Lvg DB	**64.7 / 77.0**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**1281**	CFM	Fan motor BHP	**0.70**	BHP
Design CFM/sqft	**1.58**	CFM/sqft	Fan motor kW	**0.56**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**180**	CFM	CFM/person	**6.84**	CFM/person
CFM/sqft	**0.22**	CFM/sqft			

Air System Sizing Summary for HP-112-ENGINEERING

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	HP-112-ENGINEERING		Number of zones	1	
Equipment Class	SPLT AHU		Floor Area	232.6	sqft
Air System Type	SZCAV		Location	St George Municipal, UT, USA	

Sizing Calculation Information

Calculation Months	Jan to Dec	Zone CFM Sizing	Sum of space airflow rates
Sizing Data	Calculated	Space CFM Sizing	Individual peak space loads

Central Cooling Coil Sizing Data

Total coil load	0.2	Tons	Peak coil load occurs at	July 16:00	
Total coil load	2.4	MBH	OA DB / WB	106.2 / 66.2	F
Sensible coil load	2.4	MBH	Entering DB / WB	82.9 / 62.5	F
Coil CFM at peak load	77	CFM	Leaving DB / WB	51.4 / 51.1	F
Sum of peak zone CFM	77	CFM	Resulting RH	52	%
Sensible heat ratio	1.000		Design supply temp.	55.0	F
CFM/Ton	386.0		Zone T-stat Check	1 of 1	OK
sqft/Ton	1159.9		Max zone temperature deviation	0.0	F
BTU/(hr sqft)	10.3				
Water flow @ 10.0 F rise	N/A				

Central Heating Coil Sizing Data

Max coil load	1.5	MBH	Load occurs at	Design Heating	
Coil CFM at Design Heating	77	CFM	BTU/(hr sqft)	6.5	
Max coil CFM	77	CFM	Ent. DB / Lvg DB	56.4 / 76.6	F
Water flow @ 20.0 F drop	N/A				

Supply Fan Sizing Data

Design CFM	77	CFM	Fan motor BHP	0.04	BHP
Design CFM/sqft	0.33	CFM/sqft	Fan motor kW	0.03	kW
			Fan total static	2.00	in wg

Outdoor Ventilation Air Data

Design airflow CFM	23	CFM	CFM/person	12.50	CFM/person
CFM/sqft	0.10	CFM/sqft			

Air System Information

Air System Name	**HP-STAIRS1**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**992.6**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.0**	Tons	Peak coil load occurs at	**July 13:00**	
Total coil load	**11.7**	MBH	OA DB / WB	**102.9 / 65.2**	F
Sensible coil load	**11.7**	MBH	Entering DB / WB	**73.0 / 55.2**	F
Coil CFM at peak load	**512**	CFM	Leaving DB / WB	**49.7 / 45.5**	F
Sum of peak zone CFM	**512**	CFM	Resulting RH	**33**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**523.8**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**1016.2**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**11.8**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**8.4**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**512**	CFM	BTU/(hr sqft)	**8.5**	
Max coil CFM	**512**	CFM	Ent. DB / Lvg DB	**72.0 / 88.9**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**512**	CFM	Fan motor BHP	**0.28**	BHP
Design CFM/sqft	**0.52**	CFM/sqft	Fan motor kW	**0.22**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			

Air System Sizing Summary for HP-STAIRS2

(In Alternative: Default Alternative)

Project: HAP_HamptonInn(Hilton) 05/29/2026
Prepared by: Shakespeare Engineering 1:21 PM

Air System Information

Air System Name	**HP-STAIRS2**		Number of zones	**1**	
Equipment Class	**SPLT AHU**		Floor Area	**1000.7**	sqft
Air System Type	**SZCAV**		Location	**St George Municipal, UT, USA**	

Sizing Calculation Information

Calculation Months	**Jan to Dec**	Zone CFM Sizing	**Sum of space airflow rates**
Sizing Data	**Calculated**	Space CFM Sizing	**Individual peak space loads**

Central Cooling Coil Sizing Data

Total coil load	**1.2**	Tons	Peak coil load occurs at	**July 19:00**	
Total coil load	**14.3**	MBH	OA DB / WB	**100.1 / 64.3**	F
Sensible coil load	**14.3**	MBH	Entering DB / WB	**73.0 / 55.2**	F
Coil CFM at peak load	**607**	CFM	Leaving DB / WB	**49.1 / 45.2**	F
Sum of peak zone CFM	**607**	CFM	Resulting RH	**33**	%
Sensible heat ratio	**1.000**		Design supply temp.	**55.0**	F
CFM/Ton	**509.7**		Zone T-stat Check	**1 of 1**	OK
sqft/Ton	**840.5**		Max zone temperature deviation	**0.0**	F
BTU/(hr sqft)	**14.3**				
Water flow @ 10.0 F rise	**N/A**				

Central Heating Coil Sizing Data

Max coil load	**8.6**	MBH	Load occurs at	**Design Heating**	
Coil CFM at Design Heating	**607**	CFM	BTU/(hr sqft)	**8.5**	
Max coil CFM	**607**	CFM	Ent. DB / Lvg DB	**72.0 / 86.5**	F
Water flow @ 20.0 F drop	**N/A**				

Supply Fan Sizing Data

Design CFM	**607**	CFM	Fan motor BHP	**0.33**	BHP
Design CFM/sqft	**0.61**	CFM/sqft	Fan motor kW	**0.26**	kW
			Fan total static	**2.00**	in wg

Outdoor Ventilation Air Data

Design airflow CFM	**0**	CFM	CFM/person	**0.00**	CFM/person
CFM/sqft	**0.00**	CFM/sqft			